

tiket.com: Becoming one of Indonesia's fastest- growing online travel agents with Google Cloud

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ABOUT TIKET.COM



With Google Cloud, tiket.com is controlling costs and improving scalability, reliability, performance, as well as gaining the insight to realize its vision of becoming Indonesia's most customer-centric

About tiket.com

tiket.com (<https://tiket.com/>) was established in 2011 as one of the pioneers of OTA (online travel agency) in Indonesia. The company offers flight, hotel, train, car rental, and entertainment tickets. Since tiket.com was founded, the number of employees has grown from 20 people to about 1,000 people. In 2017, tiket.com was acquired by GDP Venture, which also backed Blibli.com (<https://www.blibli.com/>). After the acquisition, tiket.com changed its logo and created a new tagline: "going anywhere? tiket.com!"

travel agent. The Google Cloud enabled tiket.com to provide quality experience to customers during peaks such as those that occur during Online Week, when air ticket and hotel stay prices increase by up to 50%.

Google Cloud results

- Provides a more personalized experience to customers
- Aligns costs with infrastructure usage and accommodates demand peaks without disruption

Gain infrastructure to build, reveal, grow, rise, user, increase, trans

Industries: Travel & Hospitality

Location: Indonesia

Private Cloud (<https://cloud.google.com/vpc/>)

aliz

About Aliz

Aliz is a Google Cloud Premier Partner that specializes in big data and machine learning. With offices in Budapest, Singapore, and Munich, the organization builds and provides solutions which help companies gain valuable insights into their business, predict the future, and optimize processes.

Google Cloud (<https://cloud.google.com/>)

Compute Engine
(<https://cloud.google.com/compute/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Virtual Private Cloud
(<https://cloud.google.com/vpc/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Dataflow (<https://cloud.google.com/dataflow/>)

- **Completes infrastructure rebuild with zero downtime**

Cloud Dataproc

(<https://cloud.google.com/dataproc/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Cloud Natural Language API

(<https://cloud.google.com/natural-language/>)

New technologies are changing travel expectations and behaviors. Travelers are now comfortable booking flights, hotels, car rentals and other products and services online. Online travel agencies (OTAs) enable customers to compare prices, read reviews, book hotels and services without hassle. Currently, tiket.com is one of the fastest growing OTAs in Indonesia. It was founded in 2011 with 20 employees. Since its establishment, the number of employees has increased to 1,000 people.

In 2017, [tiket.com](https://www.tiket.com/) (<https://www.tiket.com/>) was 100% acquired by GDP Venture, which backed Blibli.com. After the acquisition, the company drastically improved its marketing strategies. For instance, the company installed LED advertisements in airports, set up billboards around Indonesia, carried out creative social media strategies and developed co-branding with airlines. This leads to a 15x increase in revenue, a 4x rise in active users and a 4x increase in transactions over the period of 2017-2019.



Becoming the most customer-centric OTA in Indonesia

“Our vision is to become the most customer-centric OTA in Indonesia—meaning we deliver a service that provides the best possible customer experience,” says Firman Gautama, VP of Infrastructure & Security at tiket.com. “This means using technologies that are reliable, fit for purpose, and best meet our technical and business requirements, rather than focusing on bleeding-edge products and services.”

tiket.com started with an infrastructure comprising bare metal servers and virtual machines and services provided by traditional cloud services. However, as the business expanded its horizons, it needed to overcome the limitations of its existing infrastructure and enhance its data analysis and reporting capabilities. tiket.com evaluated available cloud services and selected Google Cloud (<https://cloud.google.com/>) for both infrastructure and data based on a range of considerations, including:

- **The ease of managing and analyzing data through services such as the BigQuery analytics**

data warehouse

- **The simplicity of Google Cloud infrastructure services**
- **The lower network latency of Google Cloud relative to alternative cloud services**
- **The potential to reduce its infrastructure cost**

Launch of a new Google Cloud region in Indonesia would also enable tiket.com to more easily comply with new personal data protection laws expected to be implemented in 2020.

Supporting demand spikes

Furthermore, Google Cloud could support spikes in customer demand during special events such as Online Tiket Week—when customers can obtain air tickets and hotel stays at discounts of up to 50%—without compromising availability or performance.

tiket.com established a series of objectives to optimize its Google Cloud infrastructure deployment. “We wanted to provide an improved experience for users of our website and mobile app; enable faster go-to-market for digital products, and transform our business into an ‘innovation factory,’” says Gautama. “From a technical perspective, it was easy to set up a GUI and APIs on Google Cloud; we could boot virtual machines on Compute Engine (/compute) in just 40 seconds; and Google Cloud provided an ideal starting point to use the Kubernetes open source container-orchestration system, which also gave us the flexibility to take a cloud-agnostic approach to our infrastructure.”

The business considered three options for the project: completing the exercise itself despite limited in-house experience and the need to maintain internal technology service levels; using the skills and experience of Google Cloud sales, engineering, and technical account management resources; or engaging a strategic partner that could provide industry-focused experience and “pair up” with the internal tiket.com technical team to design an architecture and deliver training in the use of Google Cloud services so tiket.com could build its end product in Google Cloud itself.

“We worked
shoulder-to-
shoulder with
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to changing

customer,
market or
business
circumstances,
and move on
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—Firman Gautama, VP
Infrastructure & Security,
tiket.com

Support from Google Cloud and partner

tiket.com opted for a combination of support from Google Cloud and a partner expert in Google Cloud technologies. After reviewing its options, tiket.com selected Google Cloud Premier Partner Aliz (<https://aliz.ai/>), a big data and machine learning specialist with offices in Budapest, Munich, and Singapore, to assist with the deployment.

The project started with a three-day deep dive exercise encompassing seven engineers from tiket.com, two engineers from Google, and one engineer from Aliz. The exercise incorporated an internal assessment of tiket.com’s technology environment, an evaluation of the business’s unwritten knowledge, and a discussion of its hopes

and dreams. The event concluded with a complete cloud and infrastructure reference architecture designed around everything-as-code—enabling tiket.com to automate repeatable tasks such as provisioning, configuration, and deployment while reducing costs, reducing risk, and deploying faster. “We worked shoulder-to-shoulder with Google Cloud and partner engineers in an everything-is-code environment that enabled us to easily build new products and features, adapt to changing customer, market, or business circumstances, and move on quickly to new projects and activities,” says Gautama.

tiket.com began simplifying its architecture, leveraging managed Google Cloud services as much as possible. “We could not just stop operating for a year to rebuild—we had to ensure zero downtime while overcoming technical challenges such as legacy devices storing several terabytes of data, servers serving more than 10 terabytes of small files, and hardware failures and disruptions,” says Gautama.

A complete rebuild

While the project started as a lift and shift to move workloads from unreliable infrastructure to Google Cloud, it ended up a complete rebuild based on infrastructure as code, continuous integration and continuous deployment flows, and full use of Compute Engine instance group management features, including autoscaling, autohealing, and load balancing. “Through the process, we learned how to use Virtual Private Cloud (/vpc), firewalls and

load balancers, both internal and external, in Google Cloud,” says Gautama. “We also learned how to adjust our continuous integration and continuous deployment and recreate our monitoring systems to support service and server recovery.”

The business completed the project in two phases. The second phase entailed deploying a number of microservices in an architecture dominated by Java, but increasingly featuring Go. The environment also incorporated technologies such as Kafka, MongoDB, Elasticsearch, SQL, and RabbitMQ. “With help from Google Cloud and Aliz, we completed the rebuild with zero downtime despite the project requiring the movement of hundreds of services, several databases, and hundreds of terabytes of data,” says Gautama.

A seamless service during demand peaks

tiket.com is achieving a range of benefits from the deployment of Google Cloud. The scalability and stability of the infrastructure ensured the business provided a seamless service over demand peaks during Online Tiket Week in early 2019. The business is also ideally positioned to support continued growth equivalent to the dramatic increases in revenue, active users, and transactions experienced between 2017 and 2019.

Furthermore, tiket.com is now better able to control and align its infrastructure costs with demand. “With Google Cloud, we do not have to pay for full capacity when it is not needed,” says Gautama. The business

plans to further improve its scalability, automation, and efficiency by deploying [Google Kubernetes Engine](#) (/kubernetes-engine) to manage its use of Kubernetes.

Data team uses internal resources

Meanwhile, the tiket.com data team opted to use internal resources to create a data architecture centered on the [BigQuery](#) (/bigquery) analytics warehouse. The team completed the project over six months from September 2018, and complements BigQuery with Kubeflow to run machine learning models on Kubernetes. Examples of machine learning models include meal predictions for airline customers, recommendation engines, and chatbot development. Using Kubeflow gives the business the flexibility during development to have multiple teams working on multiple projects at the same time.

tiket.com now accesses compute resources delivered through Compute Engine, networking functionality through VPC, and scalable load balancing through [Cloud Load Balancing](#) (/load-balancing). [Cloud Storage](#) (/storage) stores and provides easy access to data for tiket.com while [Pub/Sub](#) (/pubsub) and [Dataflow](#) (/dataflow) capture and process data. [Cloud SQL](#) (/sql) provides relational database services, while the business uses Docker running in Compute Engine for production AI/machine learning projects.

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A more personalized customer experience

tiket.com is generating a range of reports from BigQuery, primarily so management and teams can monitor business KPI results in near real time. For example, marketing teams can track online and offline marketing KPI results and allocate budget accordingly. Besides that, tiket.com is also building a recommendation engine to deliver a more personalized experience to customers, and with Natural Language API

(<https://cloud.google.com/natural-language/>), analyze

text and create sentiment analysis from user reviews, social media, and other sources.

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“We are just getting started with Google Cloud—all foundational services are in place and accelerated delivery is a reality,” says Gautama. “BigQuery and [Dataproc](#) (/dataproc) offer even greater capabilities for us to analyze data and generate the insights we need to make better business decisions and provide an even more relevant customer experience. In addition, we have a very real voice in providing feedback and feature and product recommendations to Google engineers.”



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Google Cloud

can help your

business (